

INFO6007

Project Management in IT

Week 06: Resource Management Plan

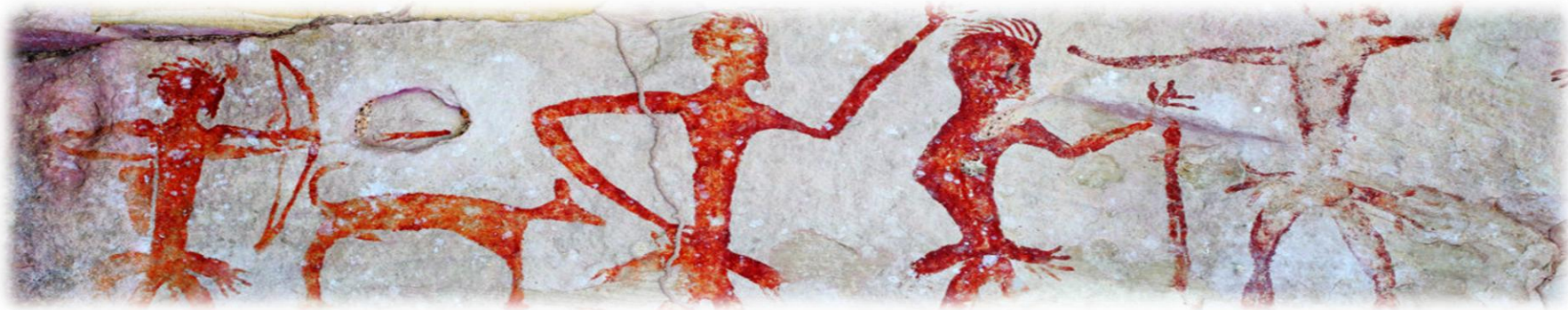
Semester 1, 2026



Acknowledgement of Country

I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. I am currently on the land of the Gadigal people of the Eora nation and pay my respects to their Elders, past, present and emerging.

I further acknowledge the Traditional Owners of the country on which you are on and pay respects to their Elders, past, present and future.



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Agenda

- Introduction
- Project Resource Management

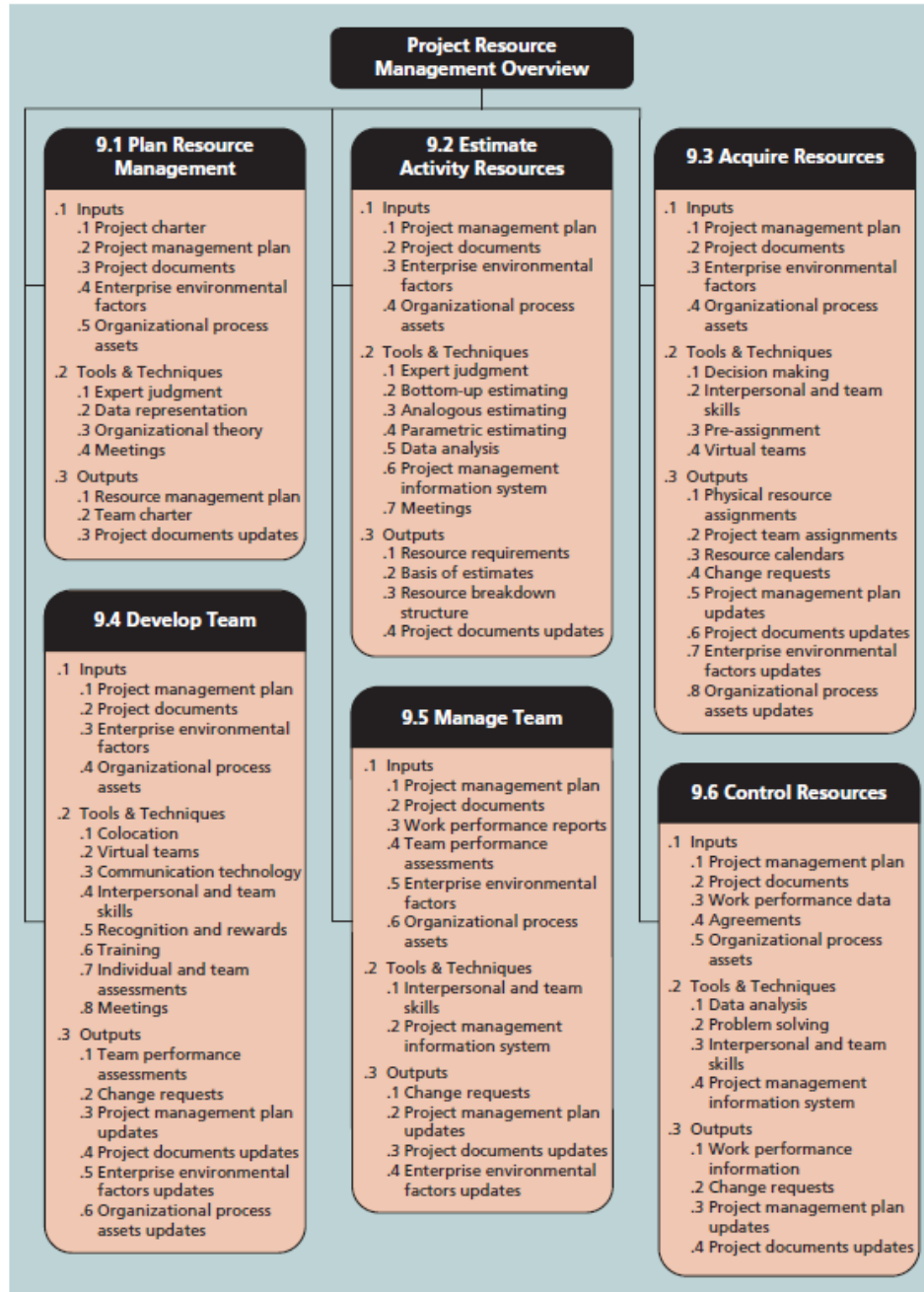
Project Resource Management

Introduction

- Project Resource Management is the process of **identifying, acquiring, and managing resources** (both human and physical) required for the successful delivery of a project.
- In IT projects, this means managing:
 - **Human resources** → Developers, testers, analysts, project managers, DevOps, data engineers, etc.
 - **Physical/technical resources** → Servers, cloud infrastructure, software licenses, hardware, tools.

Introduction

- Resource management is very important for IT projects as:
 - IT projects require specialized roles (e.g., cloud engineers, data scientists, cybersecurity experts).
 - Many resources are shared across multiple projects.
 - Licenses, cloud credits, and hardware must be planned in advance.
 - Poor planning → resource conflicts, delays, increased costs.



6 main processes in Project Resource Management

Source: *PMBOK® Guide – Sixth Edition*. Project Management Institute, Inc. (2017). Copyright and all rights reserved. Material from this publication has been reproduced with permission of PMI.

FIGURE 9-1 Project resource management overview

1. Planning Resource Management

- Planning Resource Management is the process of **defining how to estimate, acquire, manage, and use resources (human, physical, and material)** for project execution and control.

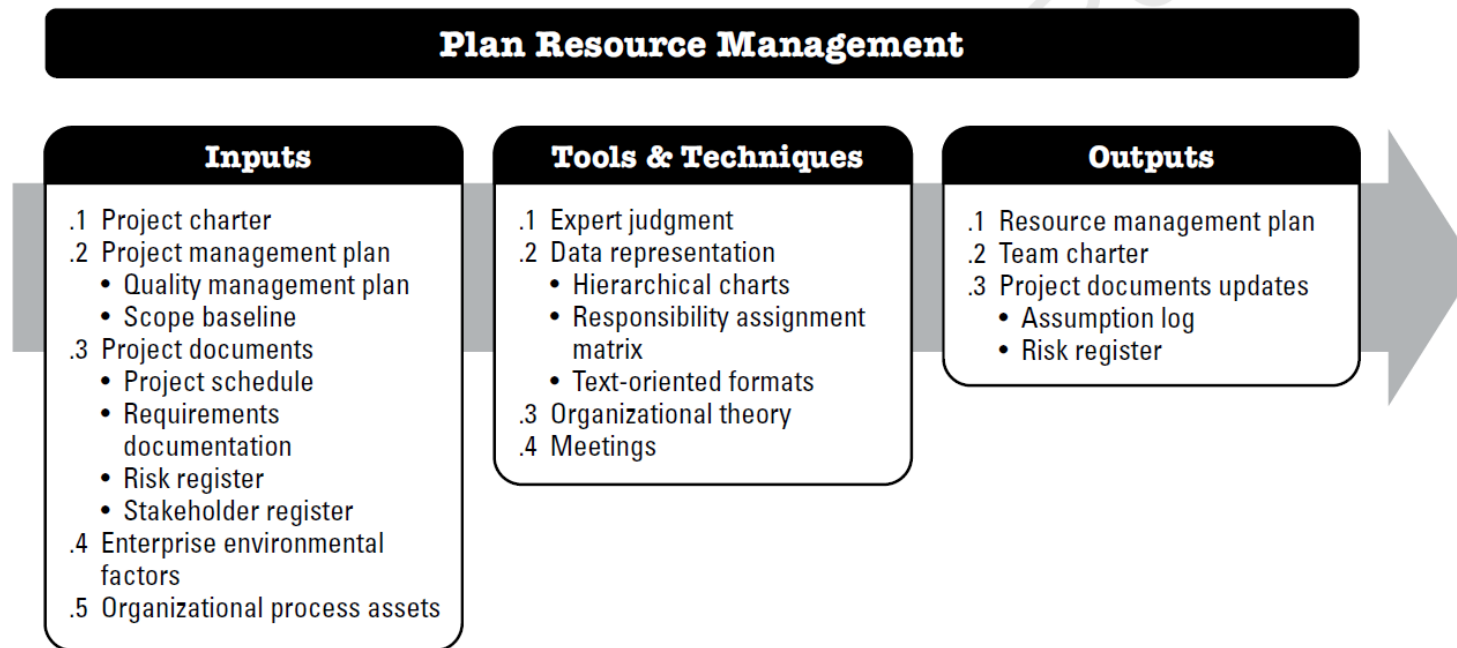


Figure 9-2. Plan Resource Management: Inputs, Tools & Techniques, and Outputs

(Roseke, 2018)

Human Resources

- Human resources are the **people involved in a project**. In IT projects, they are the most critical assets because success relies heavily on specialized skills, creativity, and collaboration.
- Human Resource Management (HRM) in projects is the process of **organizing, managing, and leading the project team**.
- The main goal is to ensure the project has the **right people with the right skills** and that they are **motivated, managed, and retained** throughout the project lifecycle.

Managing people

- People are the **most critical resource** in IT projects.
- Unlike machines or budgets, **human performance is variable**, influenced by skills, motivation, communication, and culture.
- Key aspects of managing people:
 - Leadership and motivation
 - Communication
 - Conflict Resolution
 - Performance Management
 - Team Development

Leadership in IT Projects

- **Leadership = influencing and guiding people** towards achieving project objectives.
- In IT projects, leadership is crucial because:
 - Teams are often cross-functional (developers, testers, analysts, architects).
 - Projects involve rapid technology changes.
 - Global/distributed teams need coordination.
- A good leader adapts their style to the project environment, team maturity, and organizational culture.
- There is a difference between Leadership and Management:
 - **Leadership** → Inspiring, motivating, enabling collaboration.
 - **Management** → Planning, organizing, controlling resources.

Leadership styles in IT Projects

- Autocratic (Directive)
 - Leader makes decisions unilaterally.
 - **Pros:** Fast decision-making, useful in crises (e.g., system outage).
 - **Cons:** Low motivation, stifles creativity.
- Democratic (Participative)
 - Involves the team in decision-making.
 - **Pros:** Encourages innovation, increases engagement.
 - **Cons:** Slower decisions, risk of conflict.
- Laissez-faire (Hands-off)
 - Leader provides minimal direction, gives autonomy.
 - **Pros:** Works with highly skilled, self-motivated teams.
 - **Cons:** Can lead to confusion or lack of coordination.
- And many more!

Menti - Leadership

- Imagine you are starting a new company (startup) with the aim of developing medical assistance software for NSW Health. What would be the most suitable leadership style in this scenario?



Intrinsic and Extrinsic Motivation

- **Motivation** is the internal or external drive that influences people's behavior, effort, and persistence in achieving goals.
- In IT projects, motivated team members → higher productivity, innovation, and commitment.

Aspect	Intrinsic Motivation	Extrinsic Motivation
Definition	Motivation that comes from within the individual : driven by personal satisfaction, curiosity, or passion.	Motivation that comes from external factors : rewards, recognition, or avoidance of punishment.
Characteristics	• Enjoyment of the work itself. • Desire to learn or master a skill. • Feeling of accomplishment.	• Driven by tangible or social incentives. • Performance linked to external rewards.
Source of drive	Internal (self-driven)	External (reward/punishment)
Focus	Enjoyment, mastery, curiosity	Money, recognition, promotions

Motivation Theory - Maslow's Hierarchy of Needs



- Human needs are arranged in a **hierarchy**, where **lower-level needs must be met before higher-level needs motivate behavior.**
- In IT project management, understanding these needs helps leaders **motivate and retain project team members.**

Maslow's hierarchy of needs

(Evans, 2025)

Motivation Theory - Herzberg's Motivation Hygiene Theory

- Based on studies of workplace satisfaction.
- Concluded that **job satisfaction and dissatisfaction are influenced by two separate sets of factors:**
 - **Motivators** (intrinsic, job content).
 - Relate to the work itself.
 - When present → increase job satisfaction and motivation.
 - When absent → do not necessarily cause dissatisfaction, but lead to lack of motivation.
 - **Hygiene Factors** (extrinsic, job context).
 - Relate to the work environment.
 - When absent → cause dissatisfaction.
 - When present → do not increase motivation, but prevent dissatisfaction.

Examples of Motivators and Hygiene Factors

Motivators (intrinsic, job content)

These factors make people feel satisfied, motivated, and engaged in their work.

- Achievement
- Recognition
- Responsibility
- Advancement
- Personal growth
- Meaningful or challenging work

Hygiene Factors (extrinsic, job context)

These factors do not usually increase motivation, but their absence can cause dissatisfaction.

- Salary and benefits
- Company policies
- Job security
- Working conditions
- Supervision quality
- Relationships with colleagues

Improving the effectiveness of HR on Projects

- HR in projects = **managing people** → recruiting, motivating, developing, retaining.
- Stephen Covey's **7 Habits** provide a **personal and team effectiveness framework**.
- Applying them helps project managers **improve HR processes, team engagement, and project success**.
- The 7 habits are:

Be Proactive	Begin with the end in mind	Put first things first
Think win-win	Seek first to understand, then to be understood	Synergize
Sharpen the saw		

Framework for defining and assigning work

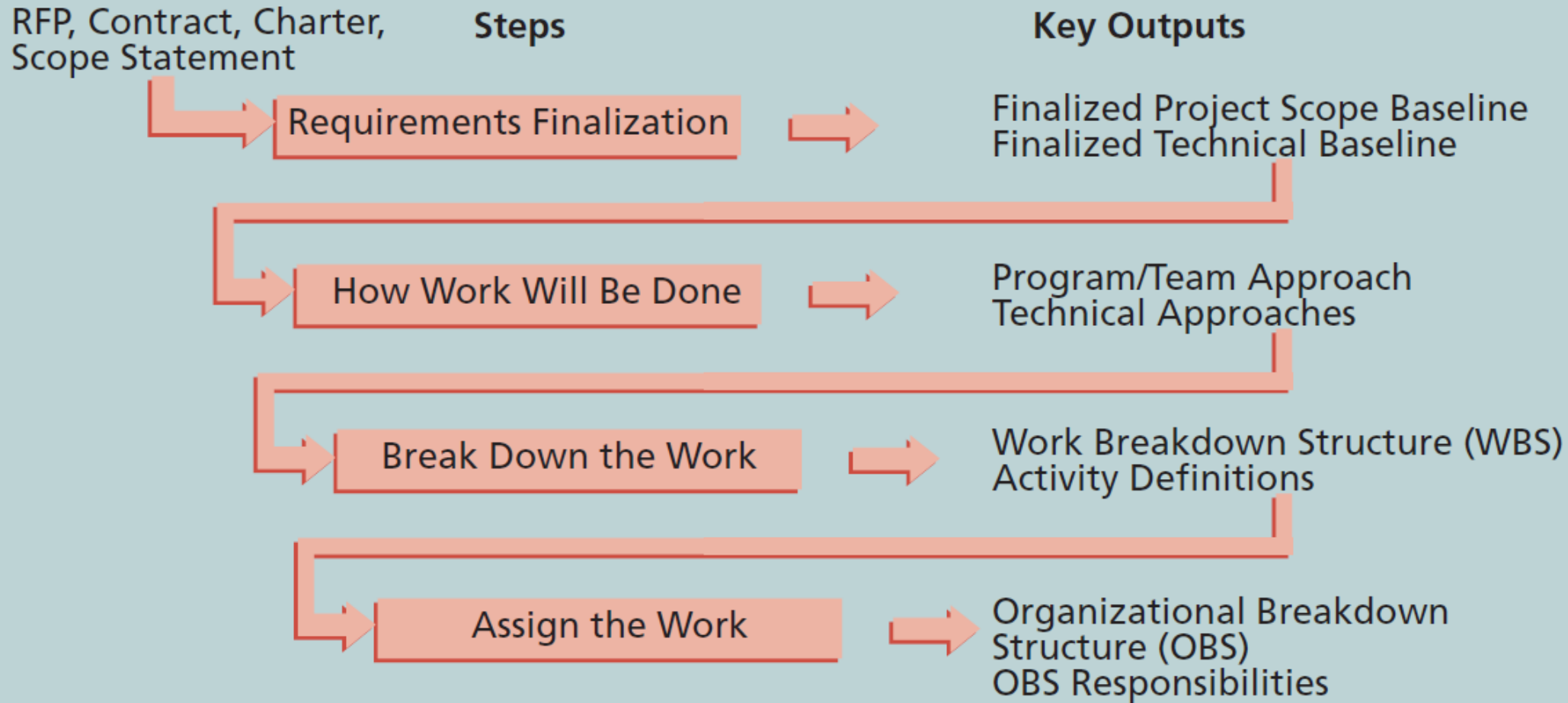


FIGURE 9-4 Work definition and assignment process

RACI Matrix

- **RACI Matrix** is a **responsibility assignment tool** that clarifies roles and responsibilities for project tasks or deliverables. Especially useful in **complex IT projects** with multiple stakeholders. Helps prevent confusion, overlaps, and gaps in accountability.
- **Responsible:** Person(s) who do the work or complete the task.
- **Accountable:** Person who is ultimately answerable; approves work and ensures task completion.
- **Consulted:** People whose input is sought before the task is done (two-way communication).
- **Informed:** People who need to be updated on progress or decisions (one-way communication).

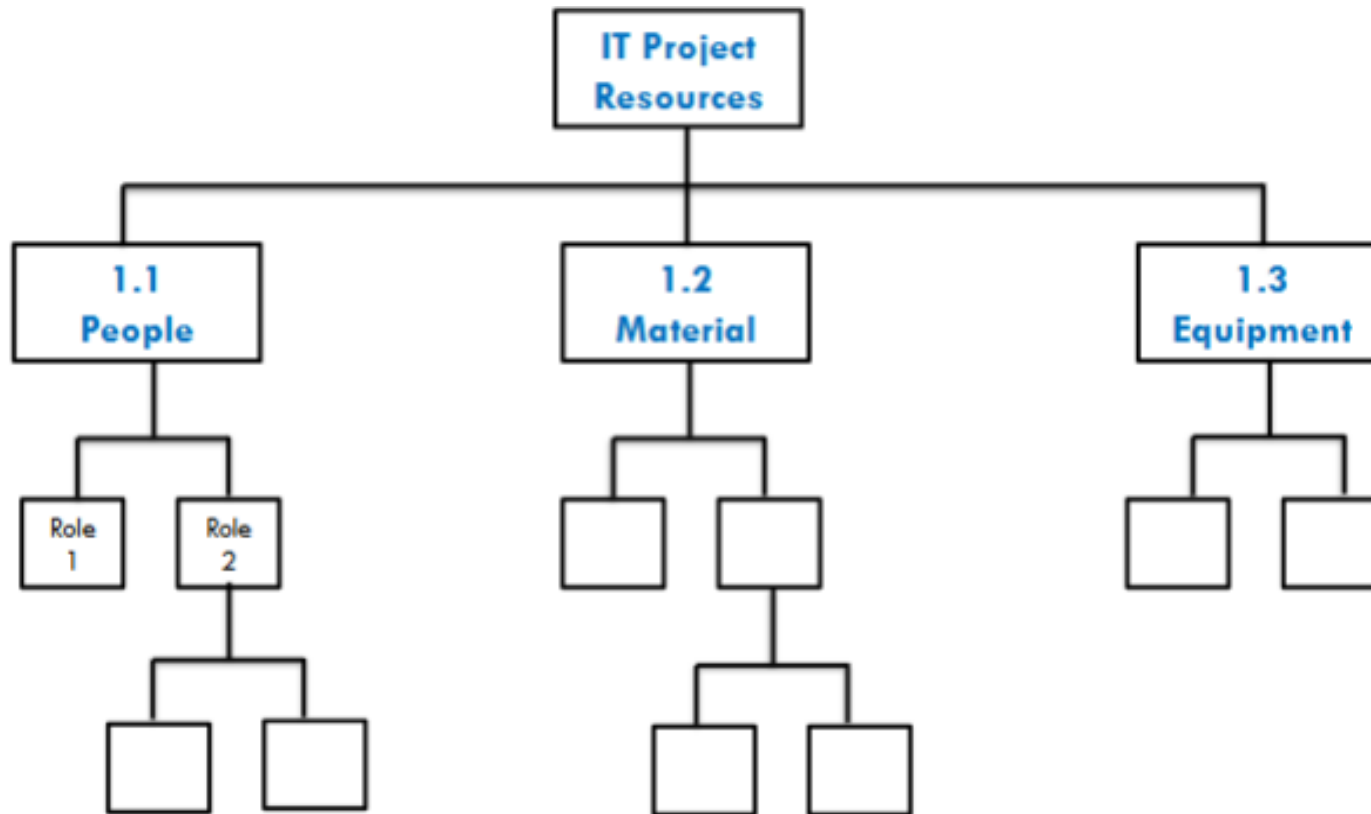
TABLE 9-2 Sample RACI chart

	Car Owner	Shop Owner	Mechanic	Parts Supplier
Pay for parts and services	A, R	C		
Determine parts and services needed	C		A, R	C
Supply parts		C	C	A, R
Install parts	I	A	R	

Resource Breakdown Structure

- A Resource Breakdown Structure (RBS) is a **hierarchical decomposition of project resources** by category and type.
- It helps in **planning, tracking, and managing all resources** (people, equipment, materials, software) in a project.
- It complements **Work Breakdown Structure (WBS)**, but focuses on **resources instead of tasks**.
- Ideally there are 3 levels of RBS:
 - **Level 1:** Broad categories of resources (Human, Equipment, Software, Materials)
 - **Level 2:** Subcategories (for humans → roles; for software → tools/platforms)
 - **Level 3:** Specific resources (names, tools, versions, servers, licenses)

Resource Breakdown Structure



Resource Histogram

- A Resource Histogram is a **bar chart that shows the amount of time each resource is allocated over a period of time.**
- It helps project managers **visualize resource usage, workload distribution, and potential over-allocation.**
- It is commonly used in **human resources planning**, but can also represent equipment, software licenses, or budget usage.
- The structure of the Resource Histogram is as follows:
 - **X-axis:** Time periods (days, weeks, months, sprints).
 - **Y-axis:** Resource usage (hours, FTEs, or number of resources).
 - **Bars:** Show how much of each resource is allocated in a given period.

Resource Histogram

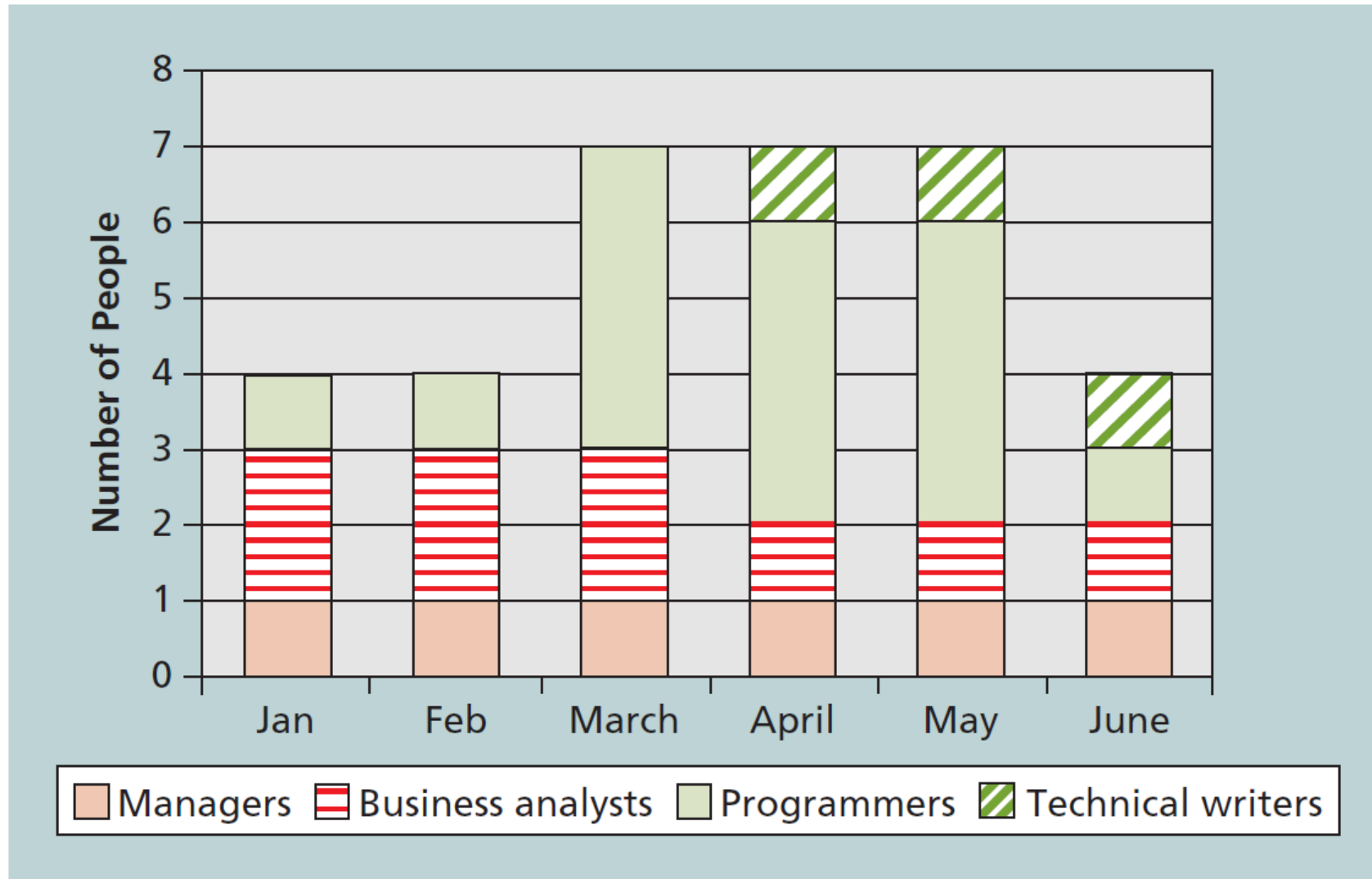


FIGURE 9-6 Sample resource histogram

2. Estimating Activity Resources

- Estimating activity resources involves **determining the types and quantities of resources** (people, equipment, materials, software) required to complete project activities.
- It is a **critical step in project planning** because accurate resource estimates affect cost, schedule, and risk management.
- In IT projects, resources may include developers, testers, cloud servers, software licenses, and tools.

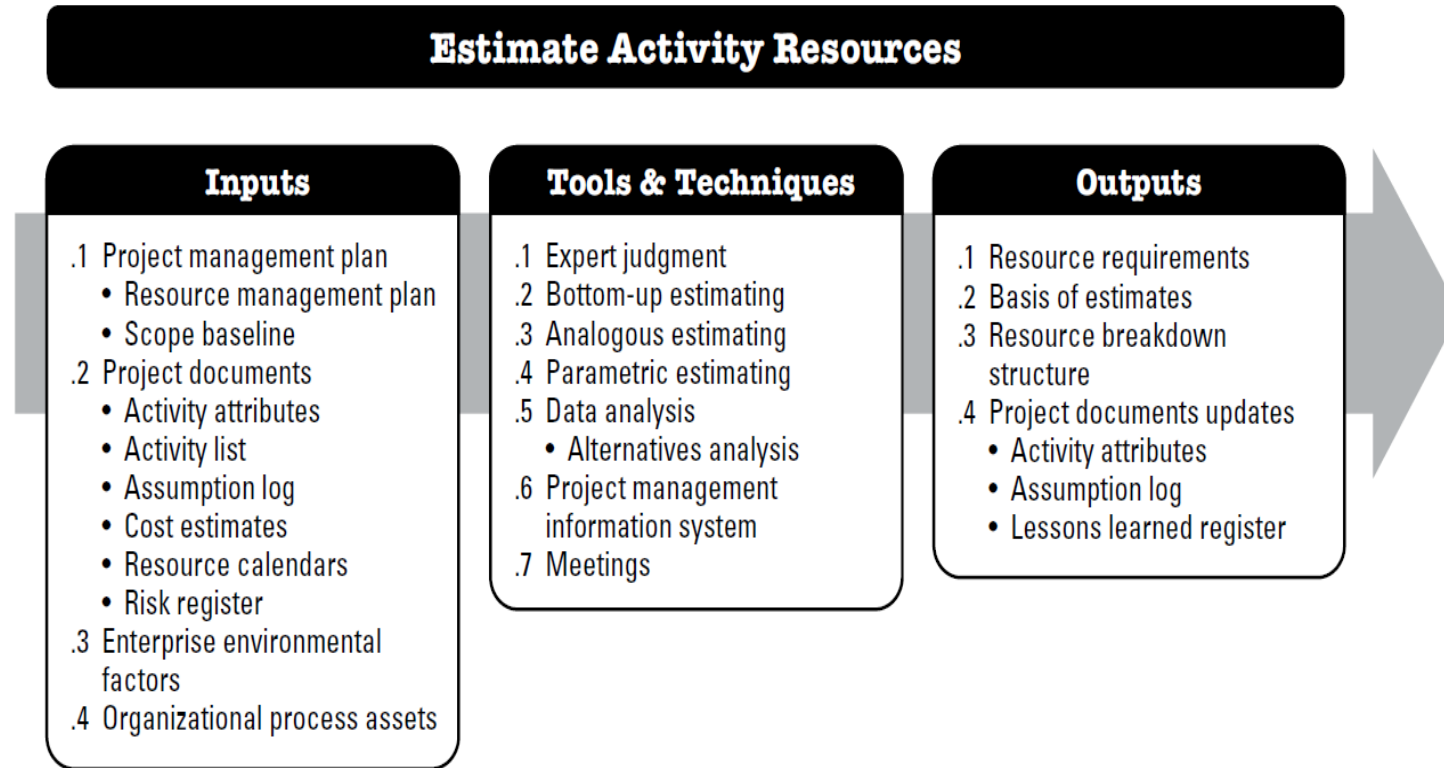


Figure 9-5. Estimate Activity Resources: Inputs, Tools & Techniques, and Outputs

2. Estimating Activity Resources

- The main purpose of this step is to:
 - Determine how many resources are needed for each activity.
 - Plan budgets and schedules accurately.
 - Identify resource constraints or gaps early.
 - Support effective workload distribution and resource leveling.
- Types of resources in IT Projects:
 - **Human Resources** – Developers, testers, designers, DevOps engineers, project managers.
 - **Equipment/Hardware** – Servers, laptops, mobile devices, networking equipment.
 - **Software/Tools** – IDEs, project management software, licenses, cloud platforms.
 - **Materials** – Documentation templates, test scripts, user manuals.

Techniques for estimating Activity resources

- Expert judgement
 - Leverage the experience of team members or subject matter experts. Useful when similar IT projects have been executed before.
- Analogous estimating
 - Base estimates on **historical data from similar projects**.
- Bottom-up estimating
 - Estimate **each activity's resources individually** → aggregate for total project. It is the most accurate but time-consuming.
- Three-point estimating (PERT)
 - Uses **optimistic (O)**, **most likely (M)**, and **pessimistic (P)** estimates.

Important questions to ask when estimating activity resources

- How difficult will specific activities be on this project?
- Is there anything unique in the project scope that will affect resources?
- Are resources available full-time or part-time?
- Can a single resource handle multiple tasks simultaneously?
- Are there potential conflicts with other projects?
- What hardware, software, or tools are required?
- Are licenses or subscriptions needed?
- Are there any constraints on availability or capacity?

3. Acquiring resources

- Resource acquisition is the process of **obtaining the human, physical, and material resources** needed to complete project activities.
- This process ensures that **the right resources are available at the right time** to prevent delays.

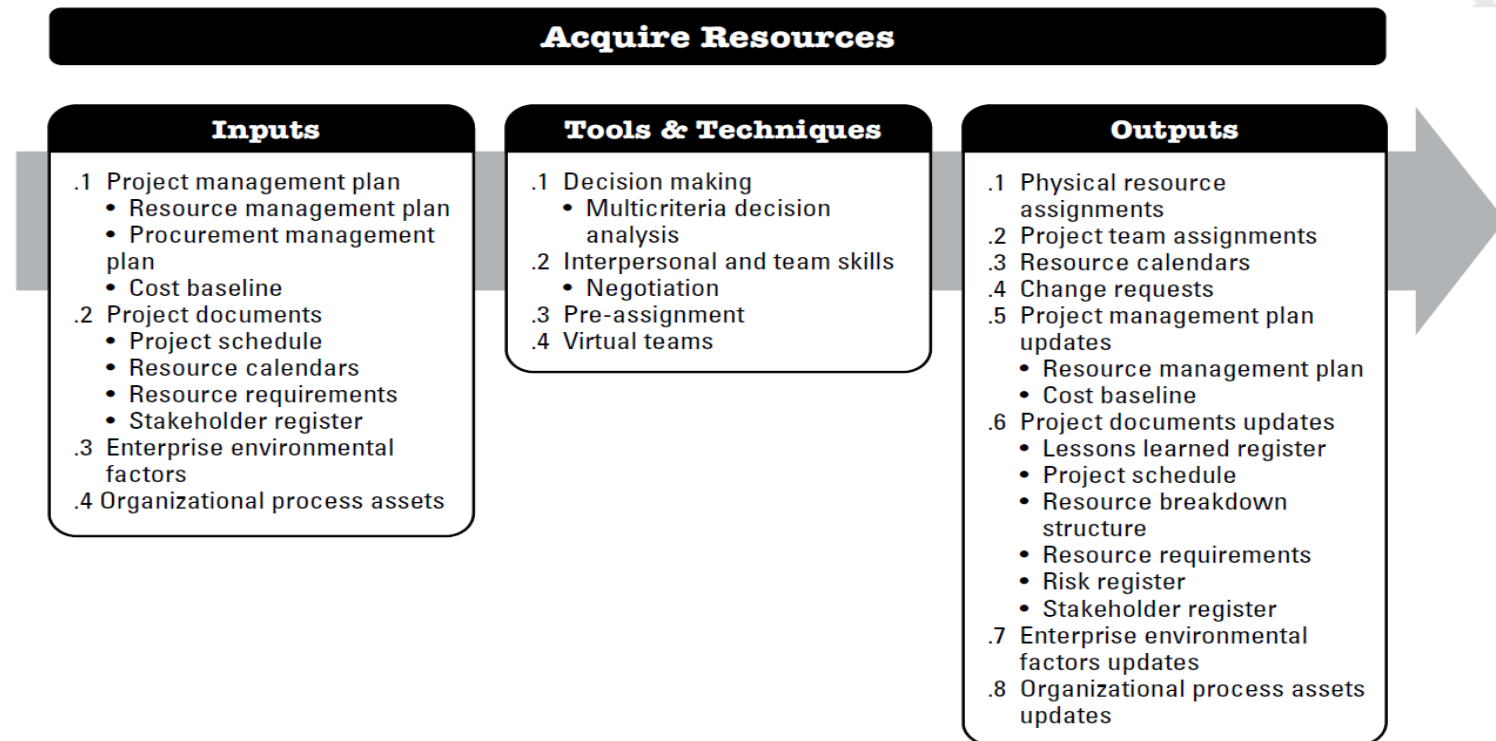


Figure 9-8. Acquire Resources: Inputs, Tools & Techniques, and Outputs

Methods of Acquiring Resources

- Internal Acquisition
 - Use existing employees or in-house resources.
 - Advantages: Familiar with company culture, faster onboarding.
- External Acquisition
 - Hire new employees, contractors, or consultants.
 - Advantages: Brings specific skills, expertise, or additional capacity.
- Procurement
 - Acquire **equipment, software, or services** through vendors.
- Multi-source Acquisition
 - Combination of internal resources, contractors, and vendors.

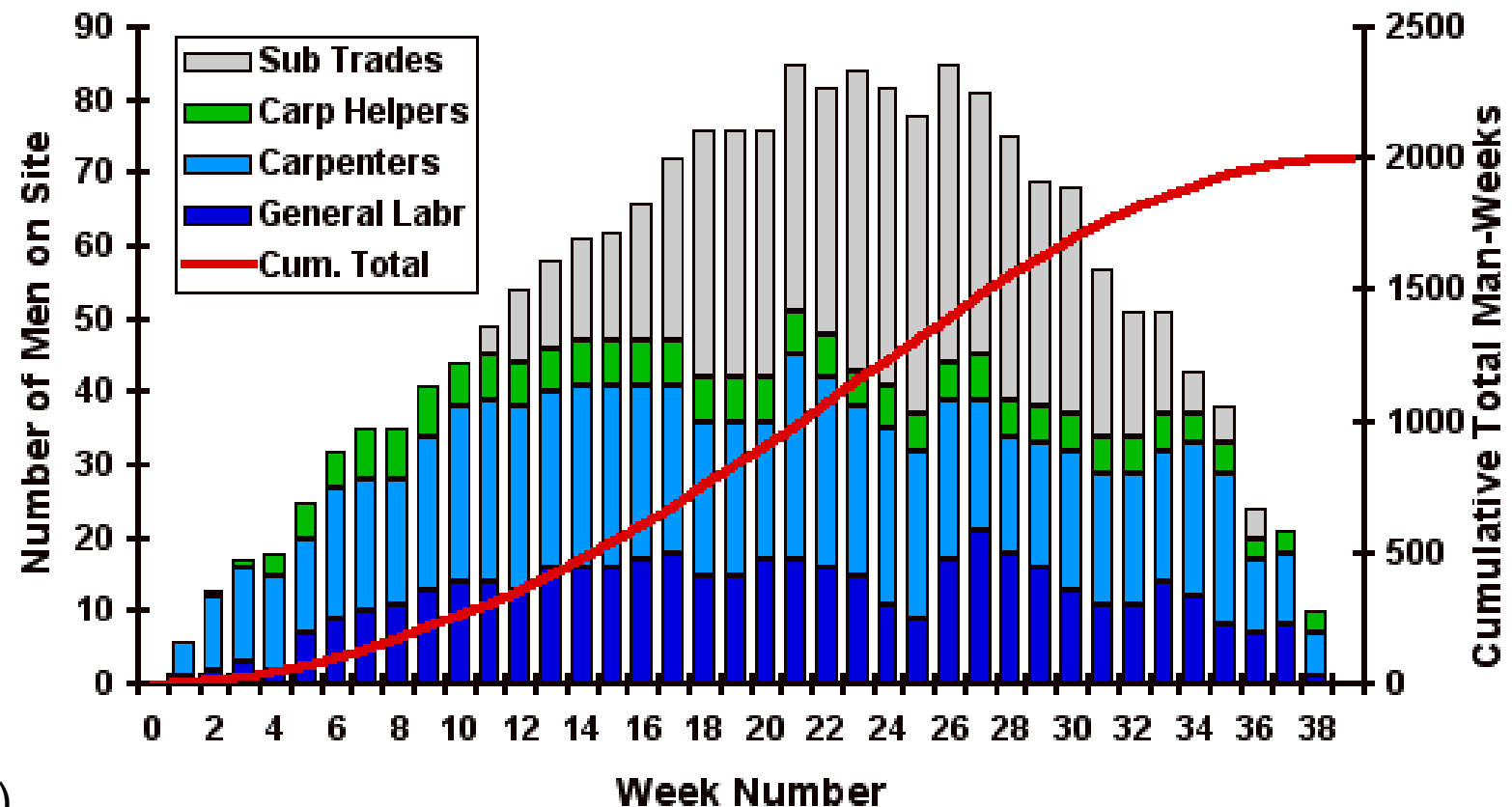
Resource Loading

- Resource loading is the process of **assigning specific resources to project activities over time** and measuring their workload.
- It helps **visualize how much work each resource has**, detect over-allocation, and optimize resource use.
- This process is closely related to **resource histograms** and **resource leveling**.
- The main purpose of this technique is to:
 - Determine how much work each resource is assigned at a given time.
 - Identify over-allocated or underutilized resources.
 - Improve schedule accuracy and productivity.
 - Support decision-making for resource adjustments or hiring.

Resource Loading

- **Units:** Hours, days, or FTE (Full-Time Equivalent) per time period.
- Formula:

$$\text{Resource Load} = \sum \text{Allocated Hours/Available Hours}$$

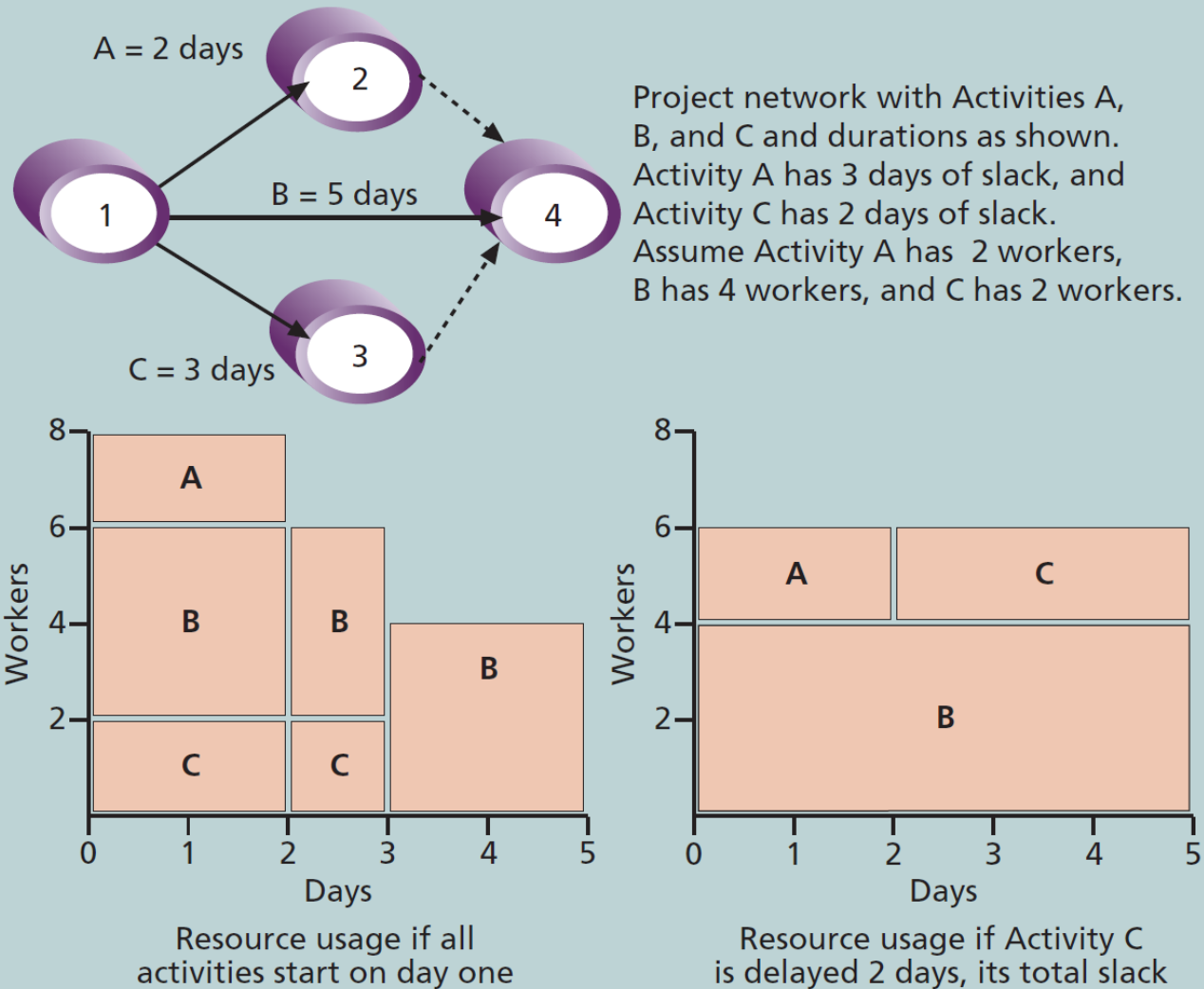


(Resource Loading, n.d.)

Resource Leveling

- Resource leveling is the process of **resolving resource over-allocation** by adjusting the project schedule or redistributing resources.
- The main goal is to ensure that **no resource is overburdened** while maintaining the project's timeline as much as possible. It is closely linked to **resource loading** and **resource histograms**.
- The main purpose of resource leveling is to:
 - Prevent burnout of team members.
 - Avoid conflicts and delays due to resource shortages.
 - Optimize resource utilization.
 - Ensure realistic project schedules.

Resource Leveling



- Techniques of resource leveling
 - Delay tasks
 - Split tasks
 - Reassign tasks
 - Adjust work hours

4. Developing the team

- Developing the project team is the process of **improving competencies, team interaction, and overall team environment** to enhance project performance.
- This step focuses on **building skills, motivation, collaboration, and trust**.
- In IT projects, team development ensures **efficient software delivery, innovation, and quality outcomes**.

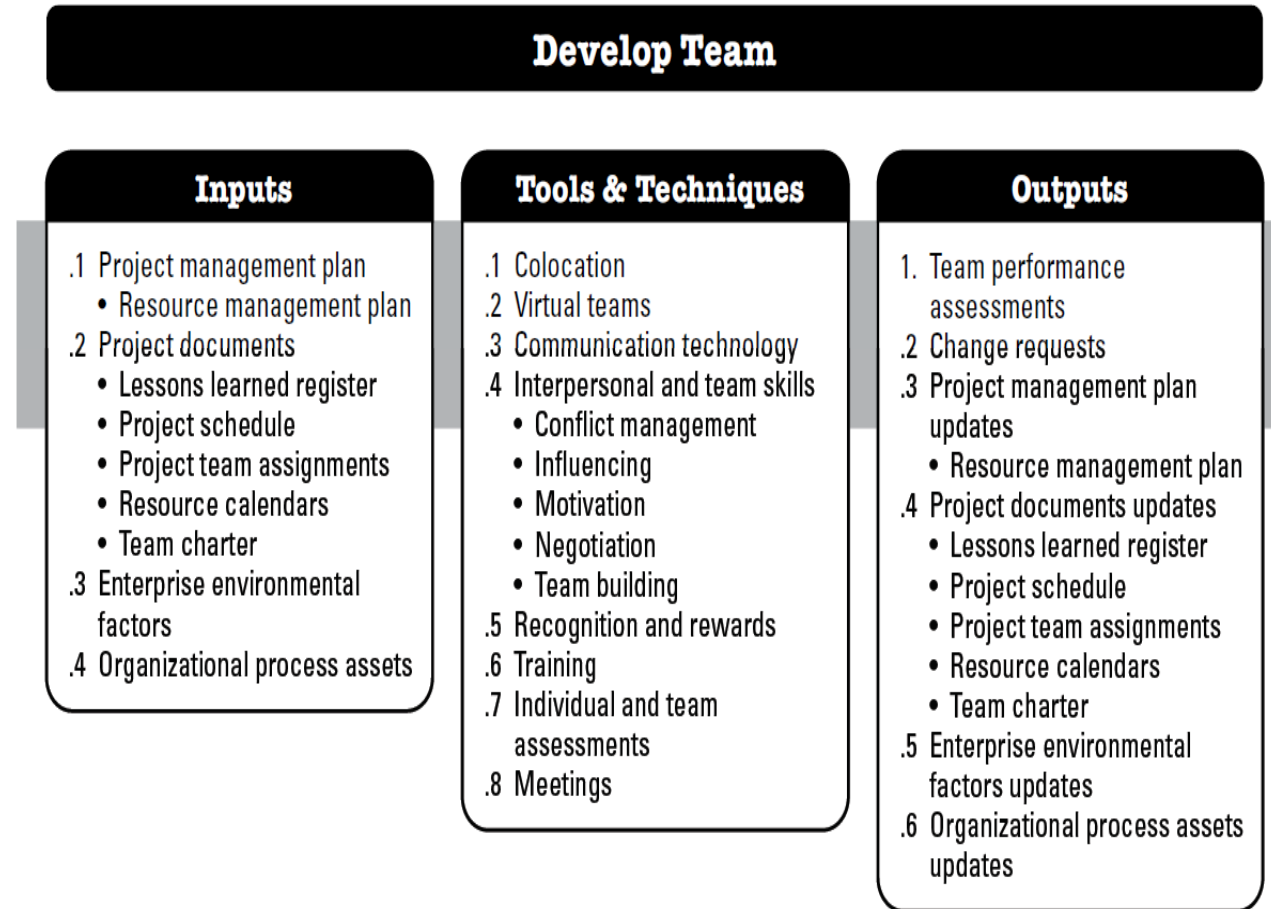


Figure 9-10. Develop Team: Inputs, Tools & Techniques, and Outputs

4. Developing the team

- The main purpose of this step is to:
 - Improve skills and competencies of team members.
 - Foster collaboration and communication.
 - Increase team motivation and engagement.
 - Enhance project performance and quality.
 - Build trust and a positive team culture.
- Key aspects of Team Development:
 - Training, mentoring, and workshops to **enhance technical and soft skills**.
 - Activities to **strengthen trust, communication, and cooperation**.
 - Recognize achievements, provide feedback, and align tasks with team interests.
 - Address interpersonal conflicts promptly to **maintain harmony and productivity**.
 - Ensure everyone **understands their responsibilities and reporting structure**.

Tuckman's Model of Team Development

- Bruce Tuckman proposed a model in 1965 describing **five stages teams go through to grow, work together effectively, and deliver results.**
- This model helps project managers **understand team behavior and guide team development.**
- It is highly relevant in IT projects where **cross-functional teams, Agile, and virtual collaboration** are common.
- The role of a project manager in developing the team:
 - Acts as a **facilitator, coach, and mediator** through all stages.
 - Monitors team dynamics, motivation, and performance.
 - Uses tools like **RACI matrix, resource loading, and skill development** to support team growth.

Tuckman's Model of Team Development

Forming

Team acquaints and establishes ground rules. Formalities are preserved and members are treated as strangers.



Storming

Members start to communicate their feelings but still view themselves as individuals rather than part of the team. They resist control by group leaders and show hostility.



Norming

People feel part of the team and realize that they can achieve work if they accept other viewpoints.



Performing

The team works in an open and trusting atmosphere where flexibility is the key and hierarchy is of little importance.



Adjourning

The team conducts an assessment of the year and implements a plan for transitioning roles and recognizing members' contributions.



Important tools and techniques for team development

- Developing a project team requires a **structured approach**. PMBOK identifies **tools and techniques** that help improve **skills, collaboration, motivation, and performance**.

Tools	Purpose
RACI Matrix	Clarify roles & responsibilities
Resource Allocation tools	Track availability and workload
Collaboration Tools	Improve communication
Performance Tracking Tools	Monitor and improve performance
Surveys/Feedback Tools	Collect team insights

Reward and Recognition systems

- Reward and recognition systems are **structured approaches to acknowledge and incentivize employees** for their contributions, achievements, and behavior.
- It helps **motivate team members, increase productivity, and improve morale**. Critical in IT projects where **team collaboration, innovation, and meeting deadlines** are key to success.
- The main purpose of these systems are to:
 - Motivate employees to perform at their best.
 - Reinforce desired behaviors and project values.
 - Increase engagement and job satisfaction.
 - Retain skilled resources and reduce turnover.
 - Encourage continuous improvement and innovation.

Types of Rewards and Recognition

- Monetary Rewards
 - Direct financial incentives for performance or achievement.
- Non-Monetary Rewards
 - Recognize contributions without direct financial compensation.
- Career based Rewards
 - Enhance employee growth and future prospects.
- Team based Rewards
 - Encourage collaboration and shared accountability.

5. Managing the team

- Managing the project team is the process of **tracking team performance, providing feedback resolving issues, and coordinating changes** to optimize project success.
- This step focuses on **guiding, monitoring, and supporting team members** throughout the project lifecycle.
- It is closely linked with **team development, resource management, and leadership**.

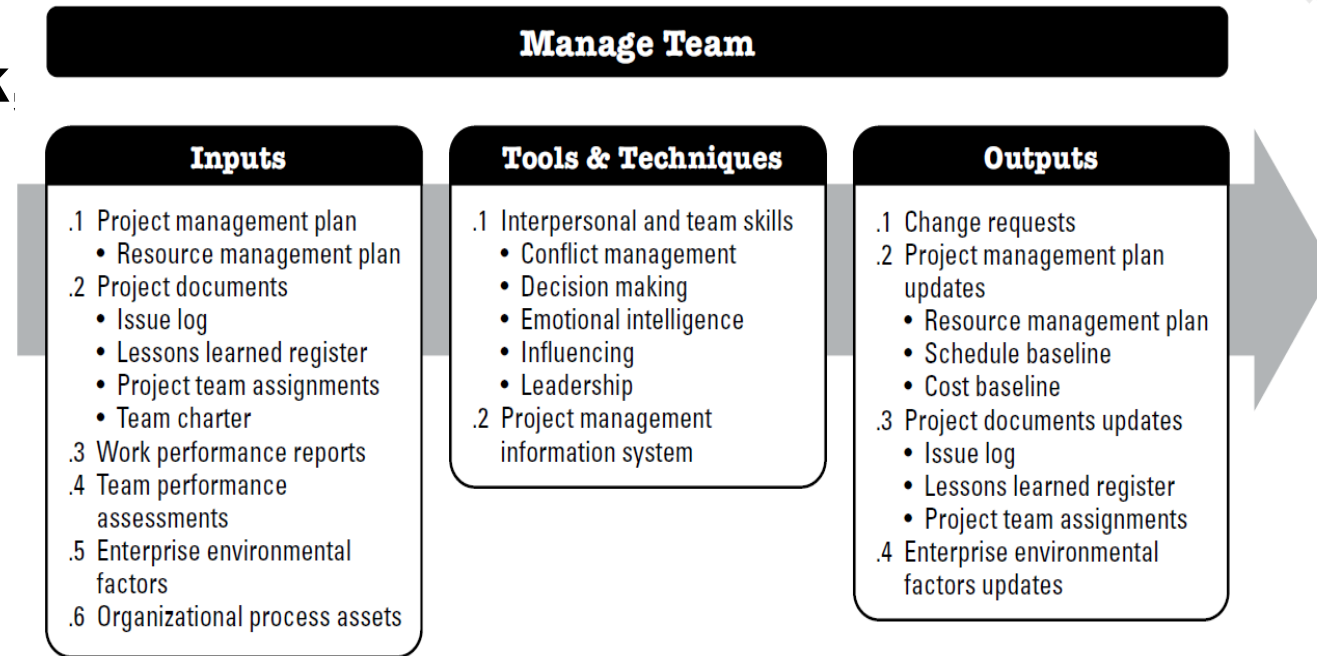


Figure 9-12. Manage Team: Inputs, Tools & Techniques, and Outputs

Managing conflicts within the team

- Conflict management is the process of **identifying, addressing, and resolving disagreements** between team members to maintain productivity and collaboration.
- Conflicts are **natural in IT projects**, due to differing opinions, priorities, and technical approaches.
- Effective conflict management ensures **team cohesion, morale, and project success**.

Relationship Importance	High	Smoothing/ Accommodating		Confrontation/ Problem-solving Collaborating
	Medium		Compromise/ Reconcile	
	Low	Withdrawal/ Avoidance		Forcing/ Directing
		Low	Medium	High
		Task Importance		

Source: Kathy Schwalbe, *An Introduction to Project Management, Sixth Edition* (September 2017)

FIGURE 9-11 Conflict handling modes

Conflict handling modes

Mode	Description	Assertiveness	Cooperativeness	When to Use
Competing	Assert own viewpoint at the expense of others (win-lose)	High	Low	Critical issues needing quick decision, limited time
Collaborating	Work with others to find a win-win solution	High	High	Complex issues requiring input from all parties
Compromising	Find middle ground (partial win-win)	Moderate	Moderate	Time-constrained situations or temporary solutions
Avoiding	Ignore or postpone the conflict	Low	Low	Minor issues or when emotions run high
Accommodating	Yield to others' needs over own	Low	High	Preserve relationships or when issue matters more to others

A dysfunctional team

- A dysfunctional team is one that **fails to work effectively together**, leading to poor performance, low morale, and project risks.
- Dysfunctional behaviors **impede collaboration, productivity, and project success**.
- It is common in IT projects due to **technical complexity, tight deadlines, remote teams, or unclear roles**.
- The 5 common causes of team dysfunction are:
 - Absence of trust
 - Fear of conflict
 - Lack of commitment
 - Avoidance of accountability
 - Inattention to results

Strategies for managing a team

- Managing a project team involves **guiding, motivating, and supporting team members** to achieve project objectives efficiently and effectively.
- Effective strategies **maximize productivity, foster collaboration, resolve conflicts, and maintain high morale.**
- A few key strategies are:
 - Define roles, responsibilities, and accountability using tools like RACI matrices.
 - Maintain **open, transparent, and regular communication.**
 - Apply an appropriate **leadership style** (transformational, servant, or situational).
 - **Identify and address conflicts early.**
 - **Acknowledge contributions** to boost motivation and morale.

6. Controlling resources

- Controlling resources is the process of **monitoring resource utilization, ensuring efficient use, resolving resource conflicts, and making adjustments** to meet project objectives.
- This step focuses on **keeping the right resources available at the right time** and preventing overallocation or underutilization.

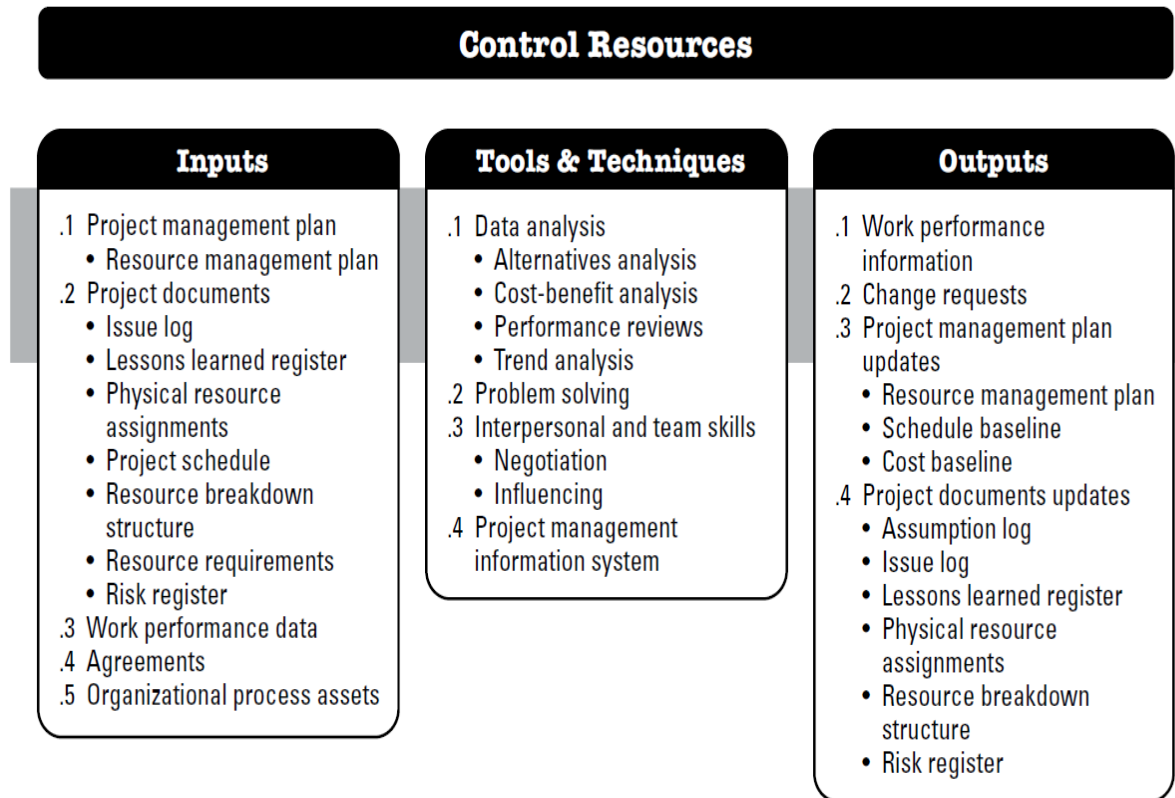


Figure 9-14. Control Resources: Inputs, Tools & Techniques, and Outputs

Understanding Productivity in Resource Management

- Productivity in resource management measures **how efficiently resources (human, equipment, or materials) are used to achieve project objectives.**
- In IT projects, productivity relates to **team output, quality of work, and efficient use of tools and technologies.**
- High productivity means **maximizing output with minimal wasted effort or resources.**

Understanding Availability in Resource Management

- Resource availability refers to the **extent to which resources (human, equipment, or materials) are accessible and ready for project tasks when needed.**
- Ensuring the right resources are available at the right time is crucial for **meeting deadlines, maintaining productivity, and avoiding project delays.**
- Availability is closely linked to **resource planning, scheduling, and allocation.**

Understanding Cost in Resource Management

- In resource management, **cost** refers to the **financial expenditure associated with acquiring, using, and maintaining resources** (human, equipment, or materials) for a project.
- Effective cost management ensures **projects stay within budget while optimizing resource utilization.**
- Cost is a **key factor in planning, acquiring, and controlling resources.**

Scenario Analysis of Productivity, Availability, and Cost

- You have the option of using any one of three team members for a project assignment. Your subject matter expert estimated that the task will take 90 effort hours.
- The following resource matrix provides individual productivity, availability, and unit cost for each of the team members.

Task	Bill			Mary			Minh		
	Prod	Avail	\$/Hr	Prod	Avail	\$/Hr	Prod	Avail	\$/Hr
Design prototype	60%	50%	\$45	120%	40%	\$80	100%	40%	\$60

- Q5: If cost is the primary consideration, who should be assigned to the task?
- Q6: If schedule is the primary consideration, who should be assigned to the task?

Scenario Analysis of Productivity, Availability, and Cost

- Cost based assignment decision

To determine the cost-effective choice:

1. **Effort required** = 90 effort hours
2. **Actual time taken** = Effort / Productivity
3. **Cost** = Time taken × \$/Hr

Calculations:

- **Bill**

$$\begin{aligned}\text{Time} &= 90 / 0.60 = 150 \text{ hours} \\ \text{Cost} &= 150 \times \$45 = \mathbf{\$6,750}\end{aligned}$$

- **Mary**

$$\begin{aligned}\text{Time} &= 90 / 1.20 = 75 \text{ hours} \\ \text{Cost} &= 75 \times \$80 = \mathbf{\$6,000}\end{aligned}$$

- **Minh**

$$\begin{aligned}\text{Time} &= 90 / 1.00 = 90 \text{ hours} \\ \text{Cost} &= 90 \times \$60 = \mathbf{\$5,400}\end{aligned}$$

- **Minh is the most cost-effective option.**

Scenario Analysis of Productivity, Availability, and Cost

- Schedule based assignment decision

To minimize time to complete:

1. Actual Time = Effort / Productivity

2. Available Time = Actual Time / Availability

Calculations:

- **Bill**

Task Time = $90 / 0.60 = 150$ hours

Calendar Time = $150 / 0.50 = \mathbf{300}$ hours

- **Mary**

Task Time = $90 / 1.20 = 75$ hours

Calendar Time = $75 / 0.40 = \mathbf{187.5}$ hours

- **Minh**

Task Time = $90 / 1.00 = 90$ hours

Calendar Time = $90 / 0.40 = \mathbf{225}$ hours

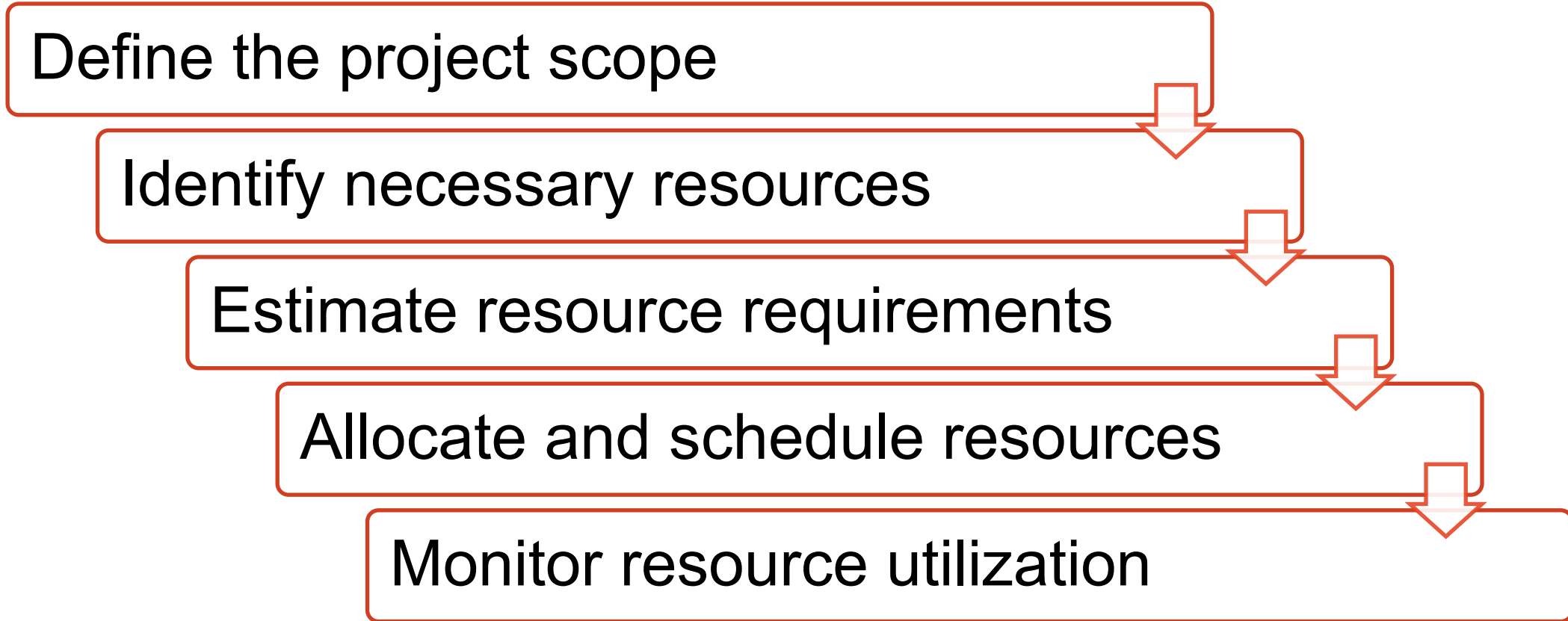
- **Mary can complete the task fastest in calendar time.**

Resource Management in Agile Environments

- In Agile, resource management is less about fixed long-term allocation and more about **matching team capacity to sprint goals**.
- Teams plan based on **available time, skills, and workload**, rather than assuming full availability.
- Agile teams are typically **cross-functional and self-managing**, so the focus is on ensuring the team collectively has the capabilities needed to deliver value in each sprint.
- This makes resource management a continuous activity, reviewed regularly as priorities and capacity change.

<https://www.atlassian.com/work-management/project-management/resource-planning>

Key Steps in Resource Management in Agile



Key Resource Management Activities in Agile

- **Capacity planning** helps the team understand how much work can realistically be completed in a sprint.
- During **sprint planning**, the team selects work by balancing business priorities with actual team capacity.
- Agile resource management also involves monitoring workload to avoid **overallocation, bottlenecks, and burnout**.
- Because Agile delivery is iterative, teams can re-balance work frequently instead of waiting for major project replans.

Best Practices in Agile Resource Management

- **Clear communication and collaboration**
 - Encourage open dialogue between team members and stakeholders to ensure everyone understands their roles and responsibilities in the resource plan.
- **Continuous monitoring and reassessment**
 - Regularly review and update your resource plan to reflect changes in project requirements or resource availability.
- **Flexibility**

Be prepared to adjust your resource plan as needed to accommodate unexpected challenges or opportunities.
- **Stakeholder involvement**
 - Involve key stakeholders in resource planning to gain insights and buy-in.
- **Documentation and reporting**
 - Maintain detailed records of your resource plan and regularly update stakeholders on resource utilization and project progress.

Managing People as the Most Important Agile Resource

- Agile places strong emphasis on **collaboration, adaptability, and shared ownership**.
- Cross-functional teams reduce handoff delays and make it easier to respond quickly to change.
- Resource management in Agile therefore includes not just staffing, but also building a team environment that supports **communication, flexibility, and sustainable delivery**.
- When done well, Agile practices can improve efficiency, reduce risk, and support continuous innovation.

<https://scrumguides.org/scrum-guide.html>

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For You!

End of Lecture Questions

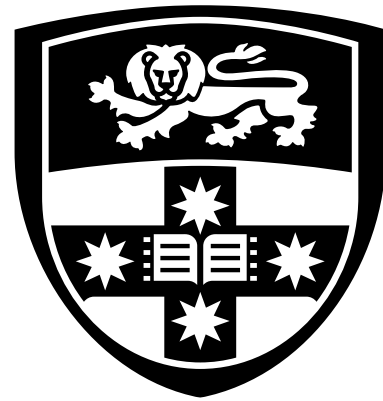
- Describe the process of **acquiring project resources** and its challenges.
- How does a **Resource Breakdown Structure (RBS)** help in resource planning?
- What tools can help **track resource allocation** effectively?
- Define **productivity, availability, and cost** in the context of project resource management.
- How can resource leveling improve both availability and cost efficiency in a project?
- Discuss the role of **automation, cross-training, and flexible scheduling** in improving resource management outcomes.

Case Study

- In a 12-week Cloud Banking Application project, the team consisted of three backend developers, two frontend developers, two QA testers, and one DevOps engineer, with a budget of \$100,000. By Week 6, resource management challenges arose: Backend Developer 1 was on leave, reducing availability and causing task delays, while the DevOps engineer struggled with complex deployments, lowering productivity. QA testers were multitasking across modules, and cloud server usage exceeded the planned budget.

Case study based questions

- Identify the **key resource management challenges** in this case study.
- Suggest **alternative strategies** to improve Backend Developer 1's productivity during leave.
- If the project budget was fixed, how would you **prioritize tasks and resources** to stay within budget?
- How could **DevOps productivity** have been monitored and optimized earlier?



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